

Executive Summary – Customer Churn Analysis

1. Project Overview

- The project focuses on analyzing customer churn.
- The dataset contains 7,043 customer records and 21 attributes.
- The main objective is to understand why customers leave a service and identify factors associated with churn.

2. Data Cleaning and Preprocessing

- Checked the dataset for missing values.
- Checked for duplicate customer IDs.
- Converted **TotalCharges** from string to numeric format.
- Blank **TotalCharges** values were replaced with zero.
- Converted **SeniorCitizen** values from 0 and 1 into Yes and No.
- Prepared the data for statistical analysis and visualization.

3. Exploratory Data Analysis (EDA)

- Analyzed customer demographics and service usage.
- Examined customer tenure, monthly charges, and total charges.
- Compared churn across different customer groups.
- Used charts and graphs to identify patterns.

4. Statistical Findings

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Metric

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Result

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| --- | --- |
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Total customers

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7,043

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Average tenure

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32.37 months

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Median tenure

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29 months

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Average monthly charges

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64.76

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Median monthly charges

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70.35

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Average total charges

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2,279.73

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Median total charges

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1,394.55

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Churn rate stated in notebook

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23.54%

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5. Customer Churn Analysis

Gender

- No substantial difference in churn was observed between male and female customers.

Senior Citizens

- Senior citizens showed a comparatively higher churn proportion.

Tenure

- Customers with shorter tenure showed higher churn.
- Customers with 1–2 months of tenure were particularly vulnerable.
- Long-term customers showed stronger retention.
- The first quartile of tenure was 9 months or less.
- The third quartile was 55 months or less.

Contract Type

- Month-to-month customers contributed higher churn counts.
- Longer-term contracts showed better retention patterns.
- The chart mainly compares customer counts, not normalized churn rates.

6. Services Analysis

The project examined:

- PhoneService
- MultipleLines

- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies

Key observations:

- Customers without OnlineBackup showed notable churn.
- Customers without TechSupport showed notable churn.
- Service usage was compared with customer churn to identify patterns.

7. Payment Method Analysis

- Electronic Check was notably associated with higher churn.
- Payment methods were compared using visualizations.
- The analysis identifies associations, not proof that a payment method causes churn.

8. Visualization and Charts

The notebook uses visualizations such as:

- Bar charts
- Stacked bar charts
- Tenure distribution charts
- Churn comparison charts
- Contract-based analysis
- Service-based analysis
- Payment-method comparisons

These charts help identify customer groups with different churn patterns.

9. Key Business Insights

- New customers need more attention during their early months.
- Month-to-month customers are an important group to monitor.
- Customers using fewer support services show notable churn patterns.
- Payment method is another useful dimension for churn analysis.
- Customer tenure is an important factor associated with retention.

10. Business Recommendations

Customer Onboarding

- Improve the onboarding experience for new customers.
- Provide guidance during the first few months.

Customer Retention

- Introduce loyalty benefits for month-to-month customers.
- Encourage longer-term customer relationships.

Service Improvement

- Promote relevant services such as OnlineBackup and TechSupport.
- Help customers understand the value of available services.

Payment Experience

- Review the electronic-check payment experience.
- Identify possible payment-related customer difficulties.

Churn Monitoring

- Monitor churn using tenure, contract type, services, and payment methods.
- Use these findings to support customer-retention planning.

11. Conclusion

The customer churn analysis identifies tenure, contract type, service usage, and payment method as important dimensions associated with customer churn.

The findings can help businesses understand customer behavior, identify groups requiring attention, improve customer experience, and develop retention strategies.

Overall objective: Reduce customer churn, improve customer satisfaction, and support long-term customer retention.